

Ariel University
Department of Computer Science
Wireless and Mobile Network Security

Wireless Protocol Investigation Lab

Evidence-Based 802.11 Security Analysis

Track 2: WPA2/WPA3 Handshake Autopsy

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Operating System: DragonOS Noble

Wireless Adapter: EDUP AX3000 WiFi 6 (USB ID: 006:ad5857)

Access Point / Router: Tenda N151 (configured as a rogue AP via *hostapd*)

Client Device: Samsung Galaxy A72

Instructor: Dr. Eyal Berliner

1. Investigative Claim

The client enforces some security-configuration parameters during the network-joining process, while tolerating changes in others. We will identify, from packet traces, where this boundary lies: whether the client proceeds, rejects, or fails during authentication, association, RSN parameter selection, SAE exchange, or EAPOL key establishment. Specifically, this investigation focuses on three security-related parameters: Protected Management Frames (PMF), Cipher Suites, and Authentication and Key Management (AKM) behavior in WPA2/WPA3 Transition Mode.

2. Evidence Plan

Evidence	Details	Purpose
Beacon / Probe Response	Advertised security capabilities	Identify the security parameters exposed by the AP before joining.
Authentication	Authentication request and response	Determine whether the client initiates authentication.
Association	Request, response, and status codes	Determine whether association succeeds or is rejected.
RSN Fields	PMF, AKM, Cipher Suites	Examine which security parameters influence client behavior.
SAE	Commit and Confirm frames	Determine whether WPA3-SAE authentication occurs.
EAPOL	4-Way Handshake messages	Determine whether key establishment begins.
Parser Output	Extracted script output	Provide reproducible evidence.
Packet Tables	Annotated packet summaries	Link conclusions to specific observations.

Table 1: Evidence plan for evaluating the investigative claim.

3. Background

This section introduces the protocol concepts required to understand the investigation and interpret the packet traces presented later in this report. Since the objective of this investigation is to determine where clients enforce or tolerate changes in security configurations during the network-joining process, the following concepts provide the necessary context for understanding the observed behavior.

3.1 The 802.11 Network-Joining Process

Before communicating over a protected wireless network, a client progresses through several stages of the joining process. The client first discovers the network through Beacon or

Probe Response frames, performs authentication, requests association, negotiates security parameters, and finally establishes encryption keys. By examining these stages, it is possible to identify the exact point at which the client proceeds, rejects, or stops during the connection attempt.

3.2 RSN Information Element

The Robust Security Network (RSN) Information Element advertises the security capabilities of the access point and provides the information required by the client to determine whether it can join the network. The RSN Information Element contains the security parameters examined in this investigation, including PMF capabilities, Authentication and Key Management (AKM) methods, and supported Cipher Suites.

3.3 Association Status Codes

Association Status Codes indicate the outcome of the association stage. A successful status code signifies that the access point accepted the client's request, whereas other status codes indicate that the request was rejected. These codes help identify whether the enforcement boundary occurs during the association process.

3.4 Security Parameters Under Investigation

This investigation focuses on three security-related parameters that may influence the client's decision to join a network.

Protected Management Frames (PMF). PMF provides protection for selected management frames against forgery and manipulation. Depending on the advertised configuration, PMF may be disabled, optional, or required.

Cipher Suites. Cipher Suites define the encryption algorithms used to protect wireless communications. Changes in the advertised cipher configuration may influence whether a client accepts or rejects the connection attempt.

Authentication and Key Management (AKM). AKM suites determine the authentication mechanism used during the joining process. In WPA3 Transition Mode, an access point may advertise both WPA2-PSK and WPA3-SAE, allowing compatible clients to select the authentication method they support.

4. Threat Model

4.1 Attacker Position

The investigation assumes an attacker located within wireless range of the client device. The attacker is capable of deploying a rogue access point that imitates a previously trusted net-

work and observes the client’s behavior during the network-joining process. The legitimate access point was used only to establish the original trusted network profile on the client device and remained powered off during the experiments.

4.2 Attacker Capabilities

The attacker can configure and operate a rogue access point that advertises modified security parameters. Specifically, the attacker can alter Protected Management Frame (PMF) settings, Cipher Suites, and Authentication and Key Management (AKM) modes presented to the client. In addition, the attacker can passively monitor 802.11 traffic using a dedicated wireless adapter operating in monitor mode.

4.3 Assumptions

The investigation assumes that the client device contains a previously saved profile of the legitimate network and attempts to reconnect automatically when a matching SSID becomes available. It is further assumed that only one security-related parameter is modified at a time, while all other experimental conditions remain as close as possible to the baseline configuration.

4.4 Attacker Limitations

The attacker does not possess any capabilities beyond those required to reproduce the authorized laboratory environment. The attacker cannot decrypt protected user traffic solely through passive observation and cannot infer information that is not exposed through the captured 802.11 frames. The attacker is limited to operating within the authorized laboratory setup and within wireless range of the participating devices. Furthermore, the attacker can only manipulate the security parameters supported by the available hardware and software infrastructure.

5. Lab Setup

The investigation was conducted in a controlled wireless laboratory environment consisting of a legitimate access point, a rogue access point, a client device, and a dedicated monitoring station. Figure 1 illustrates the experimental topology and the interactions between the participating devices.

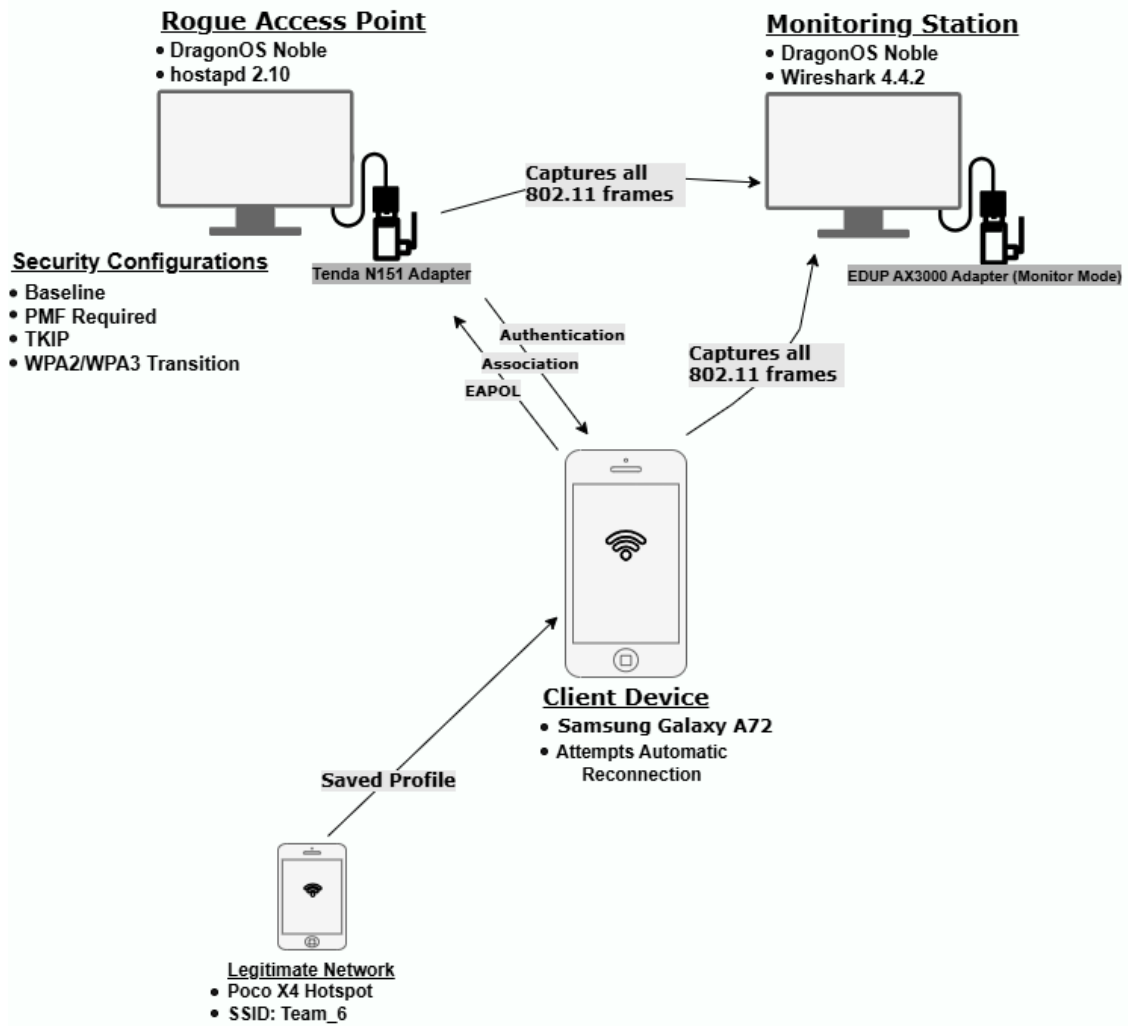


Figure 1: Experimental topology used throughout the investigation.

Legitimate Access Point

The legitimate network was provided by a Poco X4 smartphone configured as a personal hotspot. Before the experiments, the client device connected to this hotspot and stored its network profile. During the experiments themselves, the legitimate access point remained powered off.

Client Device

A Samsung Galaxy A72 smartphone served as the client device. Because the device contained a saved profile of the legitimate network, it attempted to reconnect automatically whenever a matching SSID became available.

Rogue Access Point

The rogue access point was deployed on a computer running DragonOS Noble using *hostapd* version 2.10. A Tenda N151 wireless adapter (USB ID: 002:c2e0a2) was used to advertise the same SSID as the previously trusted network while presenting the different security configurations examined throughout the investigation.

Monitoring Station

Traffic monitoring and packet analysis were performed on a separate computer running DragonOS Noble using Wireshark version 4.4.2. An EDUP AX3000 WiFi 6 wireless adapter (USB ID: 006:ad5857) operated in monitor mode and captured the exchanged 802.11 frames.

Wireless Configuration

All experiments were conducted on Channel 6 in the 2.4GHz band. The client device, rogue access point, and monitoring station remained within wireless range throughout the experiments.

Security Configurations

Table 2 summarizes the security parameters used in each experimental condition. The baseline configuration reproduced the original network settings, while each modified condition changed only a single security-related parameter.

Configuration	WPA2-PSK	SAE	CCMP	TKIP	PMF Req.	Transition
Baseline	Y	N	Y	N	N	N
PMF	Y	N	Y	N	Y	N
TKIP	Y	N	N	Y	N	N
WPA2/WPA3 Transition	Y	Y	Y	N	Y	Y

Table 2: Security parameters used in each experimental configuration.

6. Experiment Design

6.1 Baseline Condition

Before the experiments, the client device had previously connected to the legitimate wireless network and stored its network profile. During all experiments, the legitimate access point remained powered off, and only the rogue access point was activated. In the baseline condition, the rogue access point advertised the same SSID as the saved network using the original security configuration: WPA2-PSK authentication, CCMP encryption, and PMF not required. This baseline condition served as the reference configuration against which all modified conditions were compared.

6.2 Modified Conditions

Following the baseline capture, three modified conditions were examined. In all experiments, the SSID, password, wireless channel, client device, monitoring setup, and physical environment remained unchanged. Only one security-related parameter was modified at a time.

- **PMF condition:** The PMF configuration was changed from not required to required, while all other settings remained identical to the baseline condition.
- **Cipher Suite condition:** The advertised pairwise cipher suite was changed from CCMP to TKIP, while the authentication method and remaining security parameters were preserved.
- **AKM condition:** The authentication configuration was changed from a WPA2-only configuration to WPA2/WPA3 Transition Mode. In this configuration, the access point advertised both WPA2-PSK and WPA3-SAE authentication methods while also requiring PMF.

6.3 Expected Observation

Prior to collecting and analyzing the packet captures, we expected that different modifications to the advertised security configuration would influence the client’s behavior in different ways during the network-joining process.

- **PMF condition:** We expected the client to successfully reconnect despite the stricter PMF policy. Since support for Protected Management Frames is common among modern devices, we anticipated that changing PMF from not required to required would be tolerated by the client.
- **Cipher Suite condition:** We expected that changing the advertised cipher suite from CCMP to TKIP would be more likely to disrupt the reconnection attempt. Because TKIP is considered a legacy cipher and WPA2 networks typically rely on CCMP, we anticipated that the client might reject the modified configuration or fail to complete the joining process.
- **AKM condition:** We expected the client to successfully reconnect when the rogue access point advertised WPA2/WPA3 Transition Mode. Since Transition Mode is designed to maintain compatibility between devices with different authentication capabilities, we anticipated that the client would select a supported authentication mechanism and continue the connection process.

Overall, we expected that modifications intended to improve compatibility or strengthen security would likely be tolerated by the client, whereas replacing the baseline CCMP cipher with the legacy TKIP cipher would be more likely to reveal enforcement of the previously saved security configuration.

6.4 Measurement Method

Packet captures from each experimental condition were analyzed using Wireshark 4.4.2. The analysis focused on specific frame types that could indicate how far the client progressed through the network-joining process. In addition, a parser script was used to extract the same measurements from all PCAP files in a reproducible manner.

The following information was planned to be extracted from each frame type:

- **Beacon frames:** The advertised PMF setting, AKM configuration, and cipher suite were examined to identify the security parameters presented by the rogue access point in each experimental condition.
- **Authentication frames:** These frames were examined to determine whether the client initiated the connection attempt, and to distinguish between WPA2 Open System Authentication (typically 2 frames) and WPA3 SAE authentication (typically 4 frames) based on the exchange sequence.
- **Association Request and Association Response frames:** These frames were used to determine whether association occurred and to extract the association response status code, indicating whether the connection attempt was accepted or rejected.
- **EAPOL frames:** The number of EAPOL messages observed was recorded to determine whether key establishment was initiated and completed.
- **Management (Action) frames:** These frames were inspected post-connection to verify the active enforcement of Protected Management Frames (PMF). The presence of the Protected flag and encrypted payloads (CCMP parameters) served as active proof that PMF was successfully utilized.
- **Final connection outcome:** The overall result of each experiment was recorded to determine whether the client successfully reconnected under the modified security configuration.

These measurements were used to compare the baseline condition with each modified condition and evaluate how the client responded to changes in the advertised security parameters.

7. Packet Evidence

Packet-level evidence was extracted from the Wireshark captures obtained during each experiment. Representative packets were selected from the main stages of the network-joining process, including network discovery, authentication, association, and the WPA2/WPA3 4-Way Handshake. The analysis focuses on the advertised security parameters (AKM, Cipher Suite, and PMF) and their effect on the client’s ability to successfully join the network under each experimental configuration.

7.1 Baseline Configuration

The baseline experiment was performed using a rogue access point configured with the same security settings as the previously saved network profile. This capture represents the expected connection behavior and serves as the reference configuration for all subsequent experiments.

Frame	Time	Type/Subtype	Transmitter	Receiver	Important Fields	Interpretation
107	9.631381386	Probe Request	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	SSID=Team_6	The client actively searches for the previously saved network profile.
108	9.63426378	Probe Response	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	SSID=Team_6 AKM=PSK Pairwise Cipher=CCMP	The AP advertises the baseline security configuration to the client.
109	9.635540219	Authentication	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	Authentication Algorithm = Open System	The client initiates IEEE 802.11 authentication with the AP.
110	9.637405573	Authentication	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Status Code = 0 (Success)	The AP accepts the client's authentication request.
116	9.645474965	Association Request	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	SSID=Team_6 AKM=PSK Cipher=CCMP	The client requests association using the advertised baseline security configuration.
117	9.670137788	Association Response	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Status Code = 0 (Success)	The AP accepts the association request and allows the client to proceed.
118	9.70426908	EAPOL M1	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Message 1 of 4 Anonce	The AP initiates the WPA2 4-Way Handshake.
119	9.713954797	EAPOL M2	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	Message 2 of 4 SNonce MIC	The client proves possession of the Pairwise Master Key and participates in key derivation.
120	9.713975196	EAPOL M3	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Message 3 of 4 GTK Distribution MIC	The AP distributes group key material and instructs the client to install session keys.
121	9.718233904	EAPOL M4	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	Message 4 of 4 MIC	The client confirms key installation and completes the 4-Way Handshake.

Figure 2: Annotated packet table for the baseline configuration.

The packet evidence shows that the client successfully discovered the network, initiated authentication, completed association, and exchanged all four EAPOL messages required by the WPA2 4-Way Handshake.

The successful association response (Status Code 0) indicates that the access point accepted the client's association request. Furthermore, the presence of EAPOL Messages 1 through 4 demonstrates that key establishment was completed successfully.

These observations confirm that the client successfully established a connection under the baseline security configuration and provide a reference against which all modified configurations are compared.

7.2 PMF Required Configuration

In this experiment, the rogue access point advertised a modified security configuration in which Protected Management Frames (PMF) were required.

Frame	Time	Type/Subtype	Transmitter	Receiver	Important Fields	Interpretation
1	0	Probe Request	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	SSID=Team_6	The client searches for the previously saved network profile.
2	0.000011695	Probe Response	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	SSID=Team_6 PMF Required=True AKM=PSK Pairwise Cipher=CCMP	The AP advertises a modified security configuration requiring Protected Management Frames (PMF).
4	0.000015401	Authentication	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	Authentication Algorithm=Open System	The client initiates IEEE 802.11 authentication.
5	0.012068591	Authentication	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Status Code=0 (Success)	The AP accepts the client's authentication request.
6	0.012071749	Association Request	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	PMF Capable=True SSID=Team_6	The client proceeds with association and indicates support for the PMF requirement.
7	0.034623351	Association Response	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Status Code=0 (Success)	The AP accepts the association request under the PMF-required configuration.
8	0.07744594	EAPOL M1	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Message 1 of 4 Anonce	The AP initiates the WPA2 4-Way Handshake.
9	0.082419513	EAPOL M2	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	Message 2 of 4 SNonce MIC	The client participates in key derivation and proves possession of the PMK.
10	0.085401067	EAPOL M3	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Message 3 of 4 GTK Distribution MIC	The AP distributes group key material and installs session keys.
11	0.089284561	EAPOL M4	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	Message 4 of 4 MIC	The client confirms key installation and completes the 4-Way Handshake.

Figure 3: Annotated packet table for the PMF configuration.

The Probe Response packet indicates that the access point advertised PMF as a required capability. Despite this modification, the client continued through authentication and association and successfully completed the WPA2 4-Way Handshake.

Furthermore, deep packet inspection of the post-connection phase revealed the presence of Management (Action) frames with the *Protected* flag set and payloads encrypted via CCMP. This serves as active proof that the client not only accepted the PMF requirement during negotiation but actively utilized encrypted management frames to protect against forgery.

These observations confirm that the client supports PMF and successfully adapts to the stricter security requirement, actively enforcing it rather than rejecting the network.

7.3 TKIP Cipher Configuration

In this experiment, the Pairwise Cipher advertised by the access point was modified from the baseline CCMP cipher to TKIP.

Frame	Time	Type/Subtype	Transmitter	Receiver	Important Fields	Interpretation
1	0	Probe Request	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	SSID=Team_6	The client searches for the previously saved network profile.
2	0.002254019	Probe Response	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	SSID=Team_6 Pairwise Cipher=TKIP AKM=PSK	The AP advertises a modified security configuration using TKIP instead of the baseline CCMP cipher.
3	0.002267113	Authentication	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	Authentication Algorithm=Open System	The client initiates IEEE 802.11 authentication.
4	0.008939057	Authentication	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Status Code=0 (Success)	The AP accepts the client's authentication request.
5	0.008949997	Association Request	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	SSID=Team_6 AKM=PSK	The client proceeds with association despite the modified cipher configuration.
6	0.020508241	Association Response	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Status Code=0 (Success)	The AP accepts the association request.
10	0.035280492	EAPOL M1	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Message 1 of 4 Anonce	The AP initiates the 4-Way Handshake.
11	0.050989853	EAPOL M2	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	Message 2 of 4 SNonce MIC	The client proves possession of the Pairwise Master Key.
12	0.051003086	EAPOL M3	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Message 3 of 4 GTK Distribution MIC	The AP distributes group key material and installs session keys.
13	0.051005256	EAPOL M4	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	Message 4 of 4 MIC	The client confirms key installation and completes the 4-Way Handshake.

Figure 4: Annotated packet table for the TKIP cipher configuration.

The Probe Response packet confirms that TKIP was advertised as the active cipher suite. Despite this modification, the client proceeded through authentication and association without interruption.

The Association Response returned Status Code 0 and all four EAPOL messages were successfully exchanged. This indicates that the client completed the WPA/WPA2 key-establishment process and successfully joined the network.

These observations suggest that the client did not reject the network solely because of the modified cipher suite and remained capable of establishing a connection under the tested configuration.

7.4 WPA2/WPA3 Transition Configuration

In this experiment, the rogue access point advertised a WPA2/WPA3 Transition Mode configuration that supported both WPA2-PSK and WPA3-SAE authentication mechanisms.

Frame	Time	Type/Subtype	Transmitter	Receiver	Important Fields	Interpretation
161	14.97479645	Probe Request	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	SSID=Team_6	The client searches for the previously saved network profile.
162	14.98682847	Probe Response	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	SSID=Team_6 AKM=PSK+SAE Pairwise Cipher=CCMP	The AP advertises a WPA2/WPA3 transition configuration supporting both PSK and SAE authentication methods.
163	14.98949663	Authentication	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	Authentication Algorithm=SAESAE Commit	The client begins WPA3-SAE authentication.
164	15.00249379	Authentication	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Authentication Algorithm=SAE SAE Commit Response	The AP responds to the client's SAE commit.
165	15.00554815	Authentication	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	Authentication Algorithm=SAE SAE Confirm	The client confirms the SAE exchange.
166	15.00554978	Authentication	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Authentication Algorithm=SAE SAE Confirm Response	The AP completes the SAE authentication exchange.
167	15.01596417	Association Request	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	SSID=Team_6 AKM=SAE	The client proceeds with association despite the modified AKM configuration.
168	15.02977262	Association Response	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Status Code=0 (Success)	The AP accepts the association request and allows the connection process to continue.
170	15.07231142	EAPOL M1	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Message 1 of 4 Anonce	The AP initiates the 4-Way Handshake.
171	15.08815555	EAPOL M2	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	Message 2 of 4 SNonceMIC	The client proves possession of the Pairwise Master Key and participates in key derivation.
172	15.08817018	EAPOL M3	c8:3a:35:c2:e0:a2	fe:f7:ae:58:03:ea	Message 3 of 4 GTK Distribution MIC	The AP distributes group key material and installs session keys.
173	15.08817386	EAPOL M4	fe:f7:ae:58:03:ea	c8:3a:35:c2:e0:a2	Message 4 of 4 MIC	The client confirms key installation and completes the 4-Way Handshake.

Figure 5: Detailed packet sequence for WPA3-SAE authentication and EAPOL handshake.

The Probe Response packet shows the modified AKM configuration advertised by the access point. A critical observation in this configuration occurred during the authentication phase. Unlike the baseline WPA2 configuration, which utilizes a standard 2-frame Open System Authentication, this capture contained a 4-frame authentication sequence.

This 4-frame sequence confirms that the client successfully parsed the Transition Mode advertisement and actively opted to upgrade its security posture. The client executed the more robust WPA3 Simultaneous Authentication of Equals (SAE) exchange (Commit and Confirm frames) before proceeding to association.

The successful Association Response and completion of the 4-Way Handshake further indicate that the client seamlessly handled the modified AKM configuration and established a highly secure connection.

7.5 Comparative Observations

A comparison of the packet captures across all experiments reveals a consistent protocol progression. In every configuration, the client successfully completed network discovery,

authentication, association, and the WPA2/WPA3 4-Way Handshake.

Although the advertised security parameters were modified between experiments, none of the tested configurations caused the client to terminate the connection process or reject the network. However, this tolerance did not imply passivity. The packet evidence indicates a dual behavior: the client tolerated a downgrade to the deprecated TKIP cipher without alerting the user, but it also actively exploited opportunities to upgrade its security posture by switching to SAE authentication and enforcing PMF encryption when those options were available.

8. Parser / Analysis Output

Security Summary Extraction

The analysis script, named `extract_security_summary.sh`, was used to add a structured analysis layer on top of the packet evidence. While the annotated packet tables present selected frames in detail, the script summarizes each full PCAP file using the same set of security and connection-progress measurements.

- **AP security parameters:** For each capture, the script extracts the advertised Cipher Suite, AKM Suite, and PMF requirement from Beacon or Probe Response frames.
- **Client Auth Frames Count:** The script counts the Authentication frames to automatically distinguish between a standard WPA2 Open System connection (typically 2 frames) and a WPA3 SAE connection (typically 4 frames).
- **Association result:** The Association Response status code is extracted to determine whether the AP accepted the connection.
- **Key establishment:** The script counts EAPOL packets to verify that the 4-Way Handshake was completed successfully.
- **PMF Actively Used:** The script scans the post-connection traffic for Management frames with the Protected flag enabled (`wlan.fc.protected == 1`), providing automated proof of active PMF enforcement.
- **Final outcome:** The script evaluates all the above parameters to output the final connection status.

The output file, `results.csv`, contains one row for each experimental configuration. The columns include the experiment name, advertised security parameters, observed client behavior, association status, EAPOL count, and final connection status.

Experiment_Phase	AP_Cipher_Suite	AP_AKM_Suite	AP_PMF_Requirement	Client_Auth_Frames_Count	Client_Association_Status	EAPOL_Packet_Count	PMF_Observed	Final_Connection_Status
BASELINE	CCMP	PSK (WPA2)	FALSE	2 (WPA2 Open Expected)	0 (Success)	4	No	Success (Clean 4-Way)
PMF	CCMP	PSK (WPA2)	TRUE	2 (WPA2 Open Expected)	0 (Success)	4	Yes	Success (Clean 4-Way)
TKIP	TKIP	PSK (WPA2)	FALSE	2 (WPA2 Open Expected)	0 (Success)	4	No	Success (Clean 4-Way)
WPA3	CCMP	WPA3 Transition (PSK+SAE)	TRUE	4 (WPA3 SAE Expected)	0 (Success)	4	Yes	Success (Clean 4-Way)

Figure 6: Analysis script output.

The results show that all tested configurations reached the same final outcome: the client completed authentication, association, and key establishment successfully. However, the true value of the analysis script lies in revealing *how* the client reached this outcome. In the TKIP experiment, the client passively tolerated a downgraded cipher. Conversely, in the PMF and WPA2/WPA3 Transition experiments, the script actively verified that the client did not merely accept the new parameters, but dynamically adapted to them—as evidenced by the transition to a 4-frame SAE authentication sequence and the presence of encrypted Management frames.

This output provides a comparative view across all experiments, supporting the packet-level evidence and proving that modern clients dynamically balance backward compatibility with opportunistic security upgrades.

Packet Timeline Extraction

The second analysis script, named `packet_timeline.sh`, was developed to automatically extract the essential connection sequence from each packet capture and present it as a simplified protocol timeline.

- **Input and output:** The script receives a PCAPNG capture file as input and generates a CSV file containing a simplified packet timeline.
- **Tool used:** The script uses TShark to filter and extract selected IEEE 802.11 management frames and EAPOL messages from the capture.
- **Network discovery:** The script extracts the first Probe Request and the first Probe Response observed in the capture. These frames represent the network discovery stage.
- **Authentication stage:** The script extracts Authentication frames and automatically preserves either the standard WPA2 Open System authentication exchange (typically two frames) or the WPA3-SAE authentication exchange (typically four frames), depending on the observed connection behavior.
- **Association stage:** The script records the first Association Request and Association Response frames, representing the transition from authentication to network association.
- **Key establishment:** The script extracts the WPA/WPA2/WPA3 4-Way Handshake messages (EAPOL Messages 1–4) and removes duplicate retransmissions.
- **Duplicate filtering:** Retransmitted wireless frames and repeated EAPOL messages are automatically removed to ensure that the generated timeline reflects the logical protocol progression rather than capture artifacts.

The resulting CSV file contains the frame number, relative timestamp, frame type, transmitter address, and receiver address for each significant protocol event. This output provides a compact and reproducible representation of the complete connection sequence observed in each experiment.

The timeline output provides a clear visualization of how the client progressed through the network-joining process. In the Baseline, PMF, and TKIP experiments, the timeline shows the expected WPA2 sequence consisting of network discovery, a two-frame Open System Authentication exchange, association, and completion of the 4-Way Handshake. In contrast, the WPA2/WPA3 Transition experiment reveals a longer four-frame SAE authentication sequence, providing direct evidence that the client selected WPA3 authentication before continuing to association and key establishment. The timeline therefore complements the packet evidence by highlighting the exact protocol progression followed by the client in each experimental configuration. The generated CSV files were used to construct the annotated packet tables presented in Section 7. The extracted protocol timeline provided the structural framework of each table, while additional security-related fields and packet interpretations were manually added based on detailed packet inspection in Wireshark.

9. State-Machine / Sequence Diagram

The sequence diagram in Figure 7 illustrates the IEEE 802.11 protocol flow observed between the client device and the experimental access point during the network-joining process.

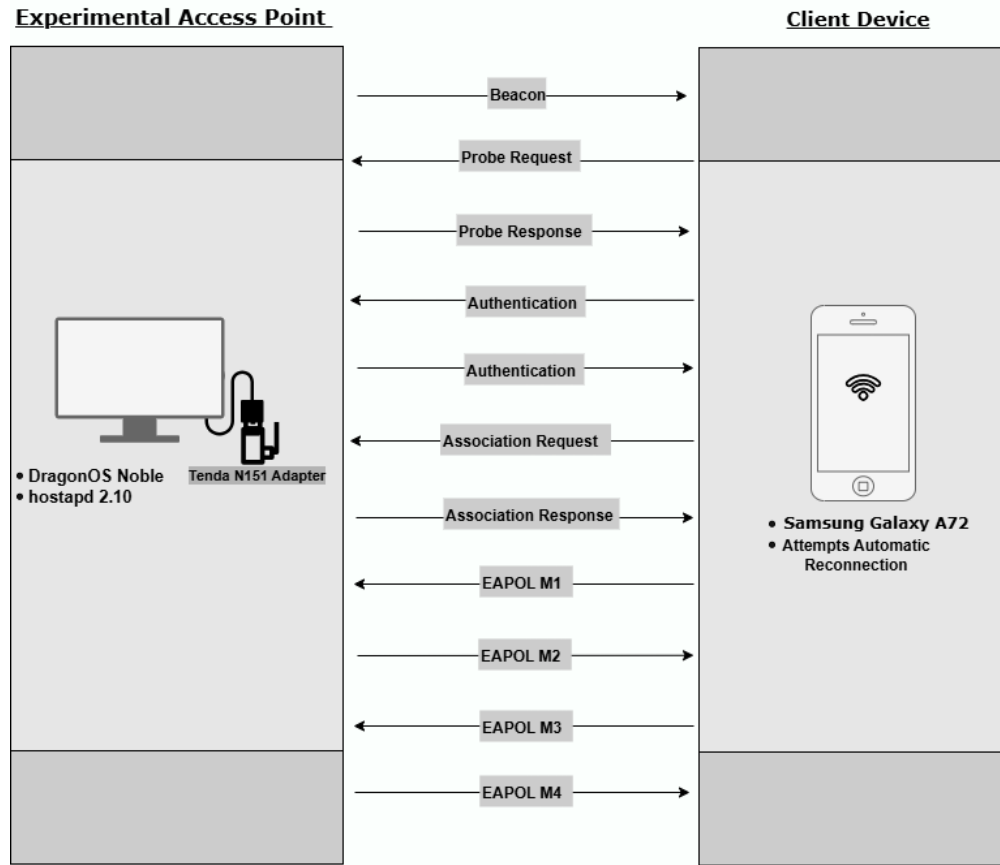


Figure 7: Protocol flow observed during the connection process.

The connection sequence is structured around four critical protocol stages, which served

as the primary checkpoints for our investigation:

1. **Network Discovery:** The client identifies the network and its advertised security capabilities (RSN, AKM, Cipher Suite, and PMF configurations) via Beacon or Probe Response frames.
2. **Authentication:** The client initiates cryptographic validation. Depending on the AP’s configuration, this involves either a standard 2-frame Open System exchange (WPA2) or a more robust 4-frame SAE Commit/Confirm sequence (WPA3 Transition Mode).
3. **Association:** The client formally requests to join the network (Association Request). A successful Association Response (Status Code 0) confirms that the AP accepted the client’s security parameters.
4. **Key Establishment:** The process concludes with the EAPOL 4-Way Handshake (Messages 1–4), deriving and installing the session keys required for secure data transmission.

By tracking the client’s progression through these specific checkpoints, we were able to systematically determine whether modifications to the advertised PMF, Cipher Suite, or AKM configurations caused the client to tolerate the changes, halt the connection process, or actively reject the network.

10. Security Analysis

10.1 Baseline Network Analysis (WPA2-PSK CCMP)

The baseline experiment observed the client’s behavior when connecting to a network whose advertised security configuration exactly matched the previously saved profile.

What the Evidence Proves

Claim	Evidence	Explanation
The AP advertises the expected WPA2 security configuration.	Frame 108 (Probe Response)	The RSN information advertises AKM=PSK and Pairwise Cipher=CCMP.
The client requests association using the advertised security parameters.	Frame 116 (Association Request)	The client negotiates AKM=PSK and Cipher=CCMP.
The AP accepts the association request.	Frame 117 (Association Response)	Status Code 0 indicates successful association.
The client successfully completes the WPA2 4-Way Handshake.	Frames 118–121 (EAPOL M1–M4)	All four EAPOL messages are observed, confirming successful key establishment.

Table 3: Claim-to-evidence mapping for the Baseline experiment.

What the Evidence Does Not Prove

The baseline experiment only demonstrates client behavior under normal conditions where the advertised security configuration matches the stored network profile. It does not provide information regarding the client’s response to security upgrades, downgrades, or configuration mismatches.

Why It Matters

This experiment establishes the reference behavior of the client. It serves as a control scenario against which all subsequent security modifications can be compared.

10.2 PMF Enforcement Analysis

The second experiment examined the client’s response to a network that required Protected Management Frames (PMF).

What the Evidence Proves

Claim	Evidence	Explanation
The AP advertises PMF as a required capability.	Frame 2 (Probe Response)	The RSN information indicates PMF Required=True.
The client accepts the PMF requirement.	Frame 6 (Association Request)	The client advertises PMF Capable=True and proceeds with association.
The AP accepts the PMF negotiation.	Frame 7 (Association Response)	Status Code 0 indicates successful association.
The client successfully establishes encrypted communication.	Frames 8–11 (EAPOL M1–M4)	The complete 4-Way Handshake is observed despite the PMF requirement.

Table 4: Claim-to-evidence mapping for the PMF experiment.

What the Evidence Does Not Prove

The experiment does not determine how a device without PMF support would behave. It also does not prove whether the client compares the PMF configuration against its stored profile before accepting the connection.

Why It Matters

The results show that the client can adapt to stronger security requirements without interrupting connectivity. This allows networks to enforce PMF protection while maintaining compatibility with supported devices.

10.3 Cipher Suite Tolerance (TKIP)

The third experiment evaluated the client’s response to a modified cipher configuration in which CCMP was replaced by TKIP.

What the Evidence Proves

Claim	Evidence	Explanation
The AP advertises a modified cipher suite.	Frame 2 (Probe Response)	The Pairwise Cipher advertised by the AP is TKIP rather than CCMP.
The client proceeds despite the cipher modification.	Frame 5 (Association Request)	The client requests association under the advertised TKIP configuration.
The AP accepts the association request.	Frame 6 (Association Response)	Status Code 0 indicates successful association.
The client successfully completes key establishment.	Frames 10–13 (EAPOL M1–M4)	The complete 4-Way Handshake is observed under the modified cipher configuration.

Table 5: Claim-to-evidence mapping for the TKIP experiment.

What the Evidence Does Not Prove

The experiment does not establish the minimum security level accepted by the client. Although TKIP was accepted, the results do not indicate whether the client would also accept weaker configurations such as WEP or an open network.

Why It Matters

The findings demonstrate that the client tolerates a legacy cipher configuration while maintaining connectivity. This behavior is relevant when evaluating downgrade scenarios involving rogue access points and Evil Twin attacks.

10.4 WPA2/WPA3 Transition Mode Analysis

The final experiment examined the client’s behavior when both WPA2-PSK and WPA3-SAE authentication mechanisms were simultaneously advertised.

What the Evidence Proves

Claim	Evidence	Explanation
The AP advertises both WPA2 and WPA3 authentication methods.	Frame 162 (Probe Response)	The RSN information advertises AKM=PSK+SAE.
The client selects WPA3-SAE authentication.	Frames 163–166 (Authentication)	The observed SAE Commit and SAE Confirm exchange indicates WPA3-SAE authentication.
The client requests association using SAE.	Frame 167 (Association Request)	The Association Request contains AKM=SAE.
The client successfully completes connection establishment.	Frames 170–173 (EAPOL M1–M4)	The complete 4-Way Handshake is observed following the SAE exchange.

Table 6: Claim-to-evidence mapping for the WPA2/WPA3 Transition experiment.

What the Evidence Does Not Prove

The experiment does not prove that the client would reject a network that advertises only WPA2-PSK. It also does not establish whether the same behavior would be observed across different client implementations.

Why It Matters

The results show that the client is capable of selecting a stronger authentication mechanism when multiple supported options are available. Understanding this behavior is important when evaluating how modern devices balance compatibility and security during the network-joining process.

11. Limitations

Although the results obtained are consistent across all experiments, several limitations should be considered:

- **Device-Specific Behavior:** The experiments were performed using a single client device (Samsung Galaxy A72). Wireless security enforcement and roaming behavior can vary significantly across different Android versions, Wi-Fi chipsets, and vendor-specific firmware implementations.
- **Controlled Environment:** The experiments were conducted in a laboratory environment with minimal interference. Real-world conditions, including high congestion, signal noise, and interference from neighboring networks, may trigger different fallback or reconnection behaviors.
- **Scope of Security Modes:** The investigation was limited to PMF, TKIP cipher suites, and WPA2/WPA3 Transition Mode. Other security-relevant configurations, such as 802.1X/EAP-based authentication or OWE (Opportunistic Wireless Encryption), were outside the scope of this work.
- **Inference Limitations:** The investigation identified whether the client connects or rejects, but it does not formally prove the client’s internal logic or the exact threshold at which it would transition from tolerance to rejection.

12. Ethics and Scope

All experimental procedures were conducted exclusively on hardware and network infrastructure owned and operated by the research group. No third-party networks, public hotspots, or unauthorized devices were targeted. Traffic analysis was strictly limited to the controlled laboratory environment. Any incidental traffic from neighboring networks detected during the monitor-mode captures was filtered out via BPF (Berkeley Packet Filter) syntax and was explicitly excluded from the analysis to ensure complete privacy protection.

13. References

1. Wireshark Foundation, *WLAN Display Filter Reference*. Available at: <https://www.wireshark.org/docs/dfref/w/wlan.html>
2. Berliner, E. (2025-2026), *Wireless and Mobile Network Security Course Materials*, Ariel University.
3. Artificial Intelligence assistance was utilized for technical consultation, diagram preparation, and documentation support during the development and analysis phases of this project.